

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.4: Trigonometry 1 — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.4: Trigonometry 1
Driving Question	DRIVING QUESTION: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance — and how does this help Kenyans solve real problems in construction, surveying, and everyday life?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Noticing the Phenomenon & Building the Need: What Tool Is Missing?	Students observed that measuring the height of a tall structure (e.g., a building under construction in Nairobi or a flag pole at school) directly is impossible without special tools. They examined right-angled triangles drawn on paper and on the ground, noting that the three sides change position depending on which acute angle, θ , is chosen as the reference angle. Teacher tip: Display a photograph of a real construction site or a tall tree and ask, 'How would a Kenyan surveyor find this height without climbing it?' Allow 2–3 minutes of think time before cold-calling 3–4 learners.	Students learned that every right-angled triangle has three named sides relative to the chosen acute angle θ : the OPPOSITE side (the side directly across from θ), the ADJACENT side (the side next to θ that is not the hypotenuse), and the HYPOTENUSE (the longest side, always opposite the 90° angle). They also learned that these labels shift when the reference angle changes — a critical conceptual anchor for all future lessons. Teacher tip: Insist learners physically point to each side and say its name aloud; mislabelling at this stage causes persistent errors throughout the sub-strand.	The three side-labels (opposite, adjacent, hypotenuse) are the vocabulary foundation on which all trigonometric ratios are built. Without precise labelling relative to a chosen angle, the ratios SOH-CAH-TOA cannot be correctly applied. This lesson establishes WHY a new mathematical tool is needed: direct measurement of heights and distances is often impractical, so we need a ratio-based method that links angles to side lengths. Connect back to the driving question: 'We now know the names of the sides — our next step is to find the ratios that stay constant no matter how big the triangle is.'
Lesson 2 Introducing the Three Trigonometric Ratios: SOH-CAH-TOA	Students measured sides of several right-angled triangles of different sizes (drawn on grid paper or measured physically with rulers) that all contained the same acute angle θ . They computed the ratio Opposite/Hypotenuse for each triangle and recorded results in a table. Teacher tip: Use at least four triangles of visibly different sizes. Ask learners, 'What do you notice about the ratio O/H each time?' Wait 30	Students learned the three fundamental trigonometric ratios for an acute angle θ in a right-angled triangle: $\sin \theta = \text{Opposite/Hypotenuse (SOH)}$, $\cos \theta = \text{Adjacent/Hypotenuse (CAH)}$, $\tan \theta = \text{Opposite/Adjacent (TOA)}$. They learned that these ratios are constant for any given angle regardless of the triangle's size — a property that makes them powerful tools for indirect measurement. Teacher tip: The	The constancy of trigonometric ratios for a fixed angle is a consequence of similar triangles: all right-angled triangles sharing the same acute angle are similar, so their corresponding side ratios are equal. This is the mathematical engine behind every height-and-distance calculation in the sub-strand. Teacher tip: Explicitly connect this to the driving question — 'Because sin, cos, and tan are always the same for a given

	seconds, then cold-call 3 learners. Repeat for A/H and O/A. The constancy of each ratio for a fixed angle is the key 'aha' moment.	mnemonic SOH-CAH-TOA should be written, spoken, and physically gestured by every learner. Cold-call at least 5 learners to state a ratio from memory before moving on.	angle, a Kenyan construction worker can use one measured angle and one known side to find any unknown side.'
Lesson 3 Sines and Cosines of Complementary Angles: $\sin \theta = \cos(90^\circ - \theta)$	Students examined pairs of acute angles in the same right-angled triangle (e.g., 35° and 55° , 40° and 50°) and compared $\sin 35^\circ$ with $\cos 55^\circ$, $\sin 40^\circ$ with $\cos 50^\circ$, using calculator values or tables. Teacher tip: Have learners complete a two-column table of $\sin \theta$ and $\cos(90^\circ - \theta)$ for $\theta = 10^\circ, 20^\circ, 30^\circ, 40^\circ, 45^\circ$. Ask, 'What pattern do you see?' Wait 1 minute before cold-calling 3 learners. Emphasise that the two acute angles in any right-angled triangle always sum to 90° .	Learners learned that in any right-angled triangle, the sine of one acute angle equals the cosine of the other acute angle: $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$. This is because the side that is 'opposite' for angle θ becomes the 'adjacent' side for the complementary angle ($90^\circ - \theta$), while the hypotenuse stays the same. Teacher tip: Use a labelled diagram showing both angles simultaneously. Reinforce with a Kenyan context: 'If a roof rafter makes a 30° angle with the horizontal, it makes a 60° angle with the vertical wall — $\sin 30^\circ = \cos 60^\circ$.'	The complementary-angle relationship halves the amount of memorisation needed: knowing $\sin$ values automatically gives $\cos$ values for complementary angles. It also foreshadows the co-function identities in higher-level mathematics. Teacher tip: Pose the reflective question, 'How does this relationship help a surveyor in Kisumu who only has a sine table but needs a cosine value?' Allow 2 minutes of pair discussion before whole-class share-out. Connect to driving question: 'Knowing this shortcut makes our calculation toolkit more flexible when working on real sites.'
Lesson 4 The Hidden Link: Discovering $\tan \theta = \sin \theta / \cos \theta$	Students computed $\sin \theta \div \cos \theta$ for several angles (e.g., $30^\circ, 45^\circ, 60^\circ$) using known or calculator values and compared the result to $\tan \theta$ for the same angles. Teacher tip: Provide a three-column table: $\sin \theta$, $\cos \theta$, $\sin \theta \div \cos \theta$. Ask, 'What do you notice about the third column compared to $\tan \theta$?' Wait 30 seconds; cold-call 3 learners. Then guide learners through the algebraic proof: $(O/H) \div (A/H) = O/A = \tan \theta$.	Learners learned that $\tan \theta = \sin \theta / \cos \theta$ for all acute angles θ . This identity follows directly from the definitions: dividing O/H by A/H gives O/A, which is exactly $\tan \theta$. They also learned that knowing any two of the three ratios allows calculation of the third — a practical tool-saving insight. Teacher tip: Stress that this is not a new rule to memorise in isolation but a logical consequence of the definitions already learned. Ask learners to derive it on their own in writing before showing the worked proof.	The identity $\tan \theta = \sin \theta / \cos \theta$ is the first algebraic identity of trigonometry and demonstrates that the three ratios are not independent — they are related. This builds algebraic reasoning skills and prepares learners for trigonometric simplification in later grades. Teacher tip: Use a Kenyan context: 'A land surveyor in the Rift Valley knows the sine and cosine of an angle from a table — they can find the tangent without a separate lookup.' Connect to driving question: 'We now have three linked tools; the next step is to know their exact values for key angles.'
Lesson 5 Trigonometric Ratios of Special Angles ($45^\circ, 30^\circ, 60^\circ, 0^\circ$, and 90°)	Students constructed a 45-45-90 triangle (by folding a square along its diagonal) and a 30-60-90 triangle (by folding an equilateral triangle in half) and measured or derived the side ratios directly. Teacher tip: Provide pre-cut squares and equilateral triangles. Ask learners to measure sides to the nearest mm and compute the ratios before deriving exact values. The hands-on construction makes the exact values memorable rather than arbitrary. Cold-call 4 learners to justify each ratio from the	Learners derived the exact trigonometric ratios: $\sin 45^\circ = \cos 45^\circ = 1/\sqrt{2}$ ($= \sqrt{2}/2$), $\tan 45^\circ = 1$; $\sin 30^\circ = \cos 60^\circ = 1/2$, $\cos 30^\circ = \sin 60^\circ = \sqrt{3}/2$, $\tan 30^\circ = 1/\sqrt{3}$ ($= \sqrt{3}/3$), $\tan 60^\circ = \sqrt{3}$; $\sin 0^\circ = 0$, $\cos 0^\circ = 1$, $\tan 0^\circ = 0$; $\sin 90^\circ = 1$, $\cos 90^\circ = 0$, $\tan 90^\circ$ is undefined. Teacher tip: Insist learners reproduce the derivation, not just memorise the table. A hand-drawn summary table in each learner's exercise book, with the diagrams beside it, is more durable than rote memorisation.	Exact values for special angles allow calculations without a calculator or trigonometric table — essential for exam settings and for quick mental estimates on a construction site. These values also recur throughout higher mathematics (vectors, waves, complex numbers). Teacher tip: Pose the question, 'Why is $\tan 90^\circ$ undefined?' ($\cos 90^\circ = 0$; division by zero is undefined.) Connect to driving question: 'When a Kenyan architect designs a 45° roof pitch or a 60° brace, these exact values

	diagram.		give precise measurements with no rounding error.'
Lesson 6 Solving Right-Angled Triangles Using Trigonometric Ratios	Students were presented with right-angled triangle diagrams drawn from Kenyan contexts (e.g., a support beam for a market stall in Mombasa, a ramp at a Nairobi hospital) and practised identifying the known side, the unknown side, and the given angle before selecting a ratio. Teacher tip: Use the three-step decision framework explicitly: (1) Label sides relative to the given angle; (2) Identify which two sides are involved (O, A, or H); (3) Choose SOH, CAH, or TOA accordingly. Require learners to write all three steps before computing. Cold-call 3 learners to explain their ratio choice before anyone calculates.	Learners learned to apply the SOH-CAH-TOA decision framework to select the correct trigonometric ratio, rearrange the equation to isolate the unknown (e.g., if $\sin \theta = O/H$, then $O = H \sin \theta$ or $H = O/\sin \theta$), and compute the answer using tables or a calculator. They also learned to find an unknown angle using inverse trigonometric functions ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) when two sides are known. Teacher tip: Common errors — using the wrong ratio because sides were mislabelled, and forgetting to rearrange before substituting. Address both explicitly with worked counter-examples.	This lesson consolidates all prior ratio knowledge into a single, repeatable problem-solving procedure. The ability to solve a right-angled triangle (find all unknown sides and angles) is the core computational skill of the sub-strand and directly answers the driving question. Teacher tip: Reinforce with a Kenyan scenario: 'A fundi (artisan) in Kisumu needs to cut a wooden rafter of unknown length for a roof with a known pitch angle and horizontal span — this is exactly the calculation they would perform.' Assign at least four varied practice problems before the next lesson.
Lesson 7 Angles of Elevation and Depression: Solving Real Height-and-Distance Problems Across Kenya	Students went outdoors (or used a classroom simulation) to measure angles of elevation to the top of the school flagpole, a tall tree, or a building using a clinometer (or a protractor-and-straw device they constructed). They recorded the angle, the horizontal distance from the object, and their eye height. Teacher tip: Prepare simple clinometers from protractors, straws, and string with a small weight. Groups of 3–4 learners each measure at two different distances to verify consistency. Ask, 'Why do we add eye height to our calculated answer?'	Learners learned that (1) the angle of elevation is measured upward from the horizontal to the line of sight to an object above the observer, and (2) the angle of depression is measured downward from the horizontal to the line of sight to an object below the observer. They learned that angles of elevation and depression between two points are equal (alternate interior angles on parallel horizontal lines), and applied $\tan \theta = \text{Opposite/Adjacent}$ to calculate unknown heights and horizontal distances in Kenyan real-world contexts (e.g., height of Kenyatta International Convention Centre, depth of a cliff at Hell's Gate, Naivasha). Teacher tip: Always draw a clearly labelled diagram — this single habit prevents the majority of errors in elevation/depression problems.	Angles of elevation and depression transform abstract trigonometry into a direct, practical measurement tool. The equality of elevation and depression angles (alternate angles) is a geometric insight that simplifies multi-step problems. Teacher tip: Connect explicitly to the driving question: 'This is the lesson where our trigonometry tool fully answers the driving question — a surveyor in the Rift Valley, a construction engineer in Nairobi, or a boat navigator on Lake Victoria can now find any height or distance using just an angle and one known measurement.' Assign a take-home problem set in a Kenyan context to cement this connection.
Lesson 8 Consolidation, Model Revision & Driving Question Resolution	Students revisited the anchoring phenomenon from Lesson 1 (finding the height of a tall structure without climbing it) and applied the full toolkit — side labelling, SOH-CAH-TOA, special angles, and elevation/depression — to solve a multi-step problem set in a Kenyan context (e.g., finding the height of a communication mast in Eldoret given an angle of elevation and a	Learners consolidated their understanding of all Sub-Strand 2.4 concepts: naming sides relative to a chosen angle, defining and applying sin, cos, and tan, using complementary-angle relationships, applying the identity $\tan \theta = \sin \theta / \cos \theta$, recalling exact values for special angles, solving right-angled triangles, and solving elevation-and-depression problems. They	The consolidation lesson closes the learning loop by returning to the original driving question with a complete answer: we CAN use one angle and one side of a right-angled triangle to find any unknown height or distance, because trigonometric ratios are constant for a given angle (a consequence of similar triangles), and we now have a systematic procedure (SOH-

	known distance). Teacher tip: Display the Driving Question Board entries from all eight lessons. Ask learners, 'Which entry from Lesson 1 can you now fully answer?' Allow 2 minutes of silent reflection before pair discussion and whole-class share-out.	revised their cumulative model to show all connections between these ideas. Teacher tip: Use a structured summary graphic (e.g., a concept map) that learners complete individually and then compare with a partner before a teacher-led class version is built on the board.	CAH-TOA with elevation/depression angles) to apply this in real Kenyan contexts — construction, surveying, navigation, and everyday problem-solving. Teacher tip: End with a metacognitive reflection prompt: 'Write one sentence explaining how your thinking about triangles changed from Lesson 1 to today.' Collect as an informal formative assessment.
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